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High-performance p-type monolayer tungsten diselenide transistors

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Two-dimensional semiconductors are potential channel materials for future scaled complementary transistor technologies due to their high carrier mobility and strong immunity to the short-channel effect. However, the electrical performance of p-type transistors is still far below that of their n-type counterparts. Here we report high-performance p-type monolayer tungsten diselenide transistors with a hole mobility of 137 $cm^2 V^{-1} s^{-1}$ and a contact resistance of approximately 560 Ω μm at room temperature. Our approach uses an industry-compatible and tunable oxygen-incorporated technique to heal the defect states. A scaled p-type monolayer tungsten diselenide transistor with a channel length of 45 nm exhibits an on-state current of 1,245 μA $μm^{-1}$ and an on/off ratio of ~10^9.

Silicon complementary metal–oxide–semiconductor (CMOS) technology will reach its performance limits at the sub-3-nm technology node[1]. Two-dimensional (2D) semiconducting transition metal dichalcogenides are a potential replacement channel material for silicon due to their dangling-bond-free surfaces, high carrier mobility at the monolayer scale and electrostatic control immunity from the short-channel effect[2,3]. To achieve further scaling of CMOS technology with 2D semiconductor materials, both n- and p-type high-performance metal–oxide–semiconductor field-effect transistors (FETs) are needed. Substantial progress has been made in 2D nFETs (using monolayer MoS_2, for example), but the performance of 2D pFETs is far from expected[4–7].

Monolayer tungsten diselenide (WSe_2) is a layered semiconductor with a direct bandgap of around 1.65 eV and a low effective mass ($0.44m_0$ for electrons and $0.41m_0$ for holes) that can also be grown over a large area by chemical vapour deposition (CVD)[8–13]. Despite its high intrinsic hole mobility, there is no reliable and practical method for fundamentally solving the hole mobility degradation problem of WSe_2 transistors because natural selenium vacancies (V_{Se}) can induce defect states and act as scatter centres to lower the carrier mobility[14–16]. Accordingly, the contact resistance (R_c) between p-type monolayer WSe_2 and metals is typically in the range of 3–200 kΩ μm due to the presence of high Schottky barriers, which greatly limits the output current of p-type WSe_2 transistors, especially in short-channel devices[17–19]. Therefore, industry-compatible approaches to passivate V_{Se} in WSe_2 films and improve the hole carrier mobility and lower the Schottky barrier height (SBH) and R_c for high-performance pFETs with a large drive current are needed.

In this Article, we report a tunable oxygen incorporation technique that can passivate V_{Se} in monolayer WSe_2 and improve the electrical performance of p-type monolayer WSe_2 transistors. The field-effect hole mobility (μ_{FE}) of monolayer oxygen-incorporated WSe_2 (O_2-WSe_2) is as high as 137 $cm^2 V^{-1} s^{-1}$ at room temperature, approximately three times greater than that of pristine WSe_2. The p-type O_2-WSe_2 transistors achieve a low SBH of around 49 meV and a low contact resistance of around 560 Ω μm due to the suppression of defect-induced states. The short-channel 45-nm monolayer O_2-WSe_2 pFET exhibits a high on-state current (I_{on}) of 1,245 μA $μm^{-1}$, which is higher than the performance of the nFET counterpart.

Monolayer WSe_2 nFETs and pFETs

Monolayer WSe_2 crystals were grown on sapphire and 100-nm SiO_2/Si substrates using low-pressure CVD in a one-inch-diameter quartz

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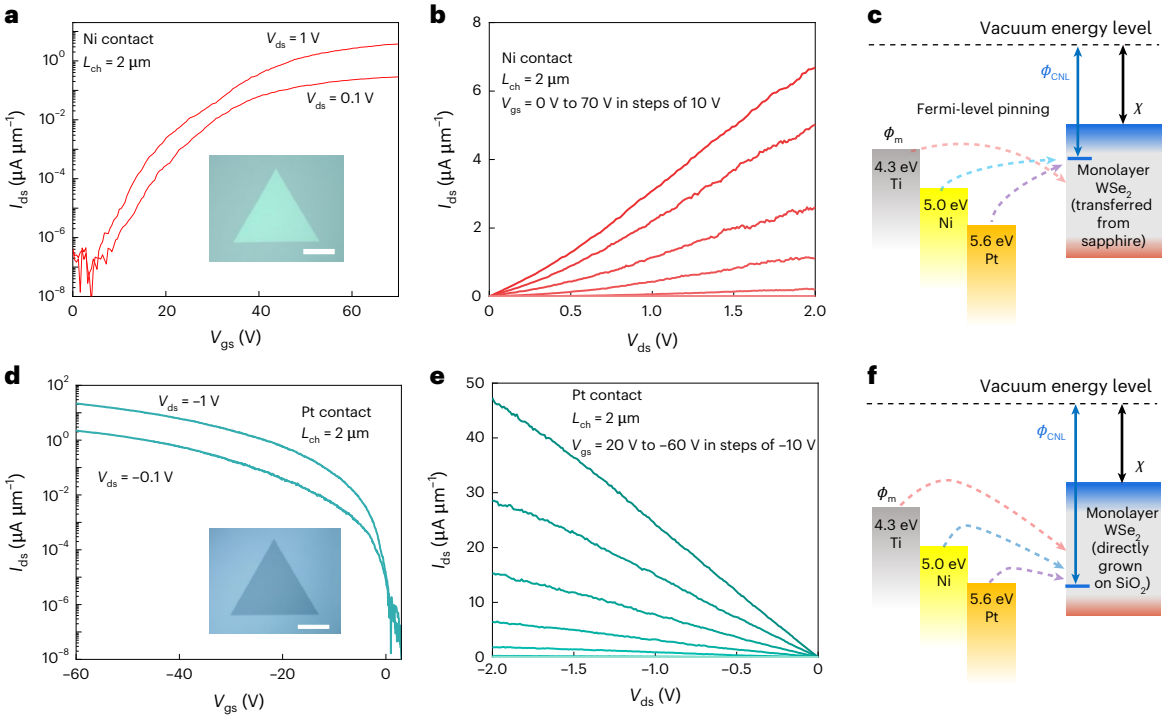


Fig. 1 | Monolayer WSe_2 nFETs and pFETs. a,**b**, Transfer (**a**) and output (**b**) characteristics of monolayer WSe_2 FETs with Ni metal contact at V_{ds} = 0.1 and 1 V. Monolayer WSe_2 is grown on sapphire substrates and then transferred to 100-nm SiO_2/Si substrates. The insets in **a** and **d** show the optical images of monolayer WSe_2 crystals grown on sapphire and SiO_2/Si substrates by CVD, respectively. Scale bar, 10 μm. **c**,**f**, Schematic of the band diagram showing the different work-function pinning positions of the contact metals to the monolayer WSe_2 transferred from sapphire substrates (**c**) and monolayer WSe_2 directly grown on SiO_2/Si (**f**). ϕ_m, work function of a metal; ϕ_{CNL}, charge neutrality level. **d**, Transfer characteristics of monolayer WSe_2 FETs directly grown on SiO_2/Si substrates. **e**, Corresponding output characteristics of the monolayer WSe_2 FETs in **d**.

tube furnace (Methods), and the typical optical images are shown in Supplementary Fig. 1[11]. For optical and electrical studies, monolayer WSe_2 films grown on sapphire were transferred onto 100-nm SiO_2/Si substrates using wet transfer approaches and monolayer WSe_2 films grown on SiO_2/Si substrates did not require a typical transfer process (Supplementary Fig. 2). The atomic force microscopy and Raman results in Supplementary Fig. 3 confirm the presence of a monolayer and high-quality WSe_2 film[8]. Figure 1a,b shows the representative transfer and output characteristics of transferred monolayer WSe_2 FETs on 100-nm SiO_2 gate dielectrics at a 2-μm channel length (L_{ch}) with a Ni metal contact. The device shows n-type transistor behaviour and a high on/off ratio of 10^8. The field-effect electron mobility of monolayer WSe_2 was 5.8 cm^2 V^{-1} s^{-1} ($\mu_{FE} = g_m L/(WC_{ox}V_{ds})$, where g_m is the transconductance; W and L are the channel width and length, respectively; V_{ds} is the source–drain voltage; and C_{ox} = 0.034 μF cm^{-2} is the gate capacitance). In particular, the electron mobility can be improved twofold via the use of high-k dielectrics (Supplementary Fig. 4) due to dielectric screening[20]. Using the epitaxial growth of high-quality films and semimetal contacts (such as Bi or Sb) and soft metals (such as In), the carrier mobility and electrical performance of monolayer WSe_2 nFETs are expected to further improve[4,21,22]. In addition, when Ti and Pt metal contacts are deposited by the common method of electron-beam evaporation, the monolayer WSe_2 transistor exhibited consistent n-type transistor behaviour due to Fermi-level pinning (Supplementary Fig. 5). To prove the Fermi pinning position, we measured the temperature-dependent electrical properties to extract the SBHs for the transferred WSe_2 transistors with different contact materials (Supplementary Fig. 6). Both low- and high-work-function metals are aligned near the conduction band of the transferred monolayer WSe_2, resulting in monotonous n-type electrical characteristics (Fig. 1c). We also fabricated transistors based on monolayer WSe_2 directly grown on 100-nm SiO_2/Si substrates using a global gated structure with three kinds of contact metal, including the low-work-function metal Ti and the high-work-function metals Ni and Pt. The transfer characteristics of monolayer WSe_2 FETs with Ti and Ni metal contacts exhibit ambipolar and p-type transistor behaviours, respectively (Supplementary Fig. 7). Figure 1d shows the transfer characteristics of monolayer WSe_2 FETs using Pt metal contacts, which also clearly exhibit p-type transistor behaviour and a high on/off ratio of ~10^8. The field-effect hole mobility of monolayer WSe_2 was 45.2 cm^2 V^{-1} s^{-1}. The corresponding output characteristics are shown in Fig. 1e. The 2-μm-long WSe_2 device has a drain current I_{ds} of ~50 μA μm^{-1} at V_{ds} = −2 V. The corresponding SBHs for the p-type WSe_2 transistors are shown in Supplementary Fig. 8. Figure 1f shows a scheme of the band diagram of the different work-function pinning positions of the contact metals to monolayer WSe_2. Therefore, monolayer WSe_2 grown on sapphire and SiO_2/Si substrates shows opposite transport types. Using n- and p-type WSe_2, we demonstrated a WSe_2-based CMOS circuit—an inverter (Supplementary Fig. 9).

Gate-dependent photoluminescence measurements at 3.4 K

We perform low-temperature photoluminescence (PL) measurements to investigate the defect states in two types of monolayer WSe_2. The first type of material was originally grown on sapphire and then transferred to SiO_2/Si substrates (transferred WSe_2), and the other type of material was grown directly on SiO_2/Si substrates (as-grown WSe_2), as shown in Fig. 2 (also shown in Supplementary Fig. 10). With decreasing temperature to 20 K, the resonance energies of free excitons blueshift for both samples, which is in agreement with previous studies[23]. Additionally, both samples exhibit a broad PL emission peak at lower energies, with the intensity increasing as the temperature decreases. This is attributed to the defect-bound excitons X_D[24]. To gain a better understanding of the exciton states in WSe_2, we conducted gate-dependent PL measurements at 3.4 K. The PL spectra of both types of sample at

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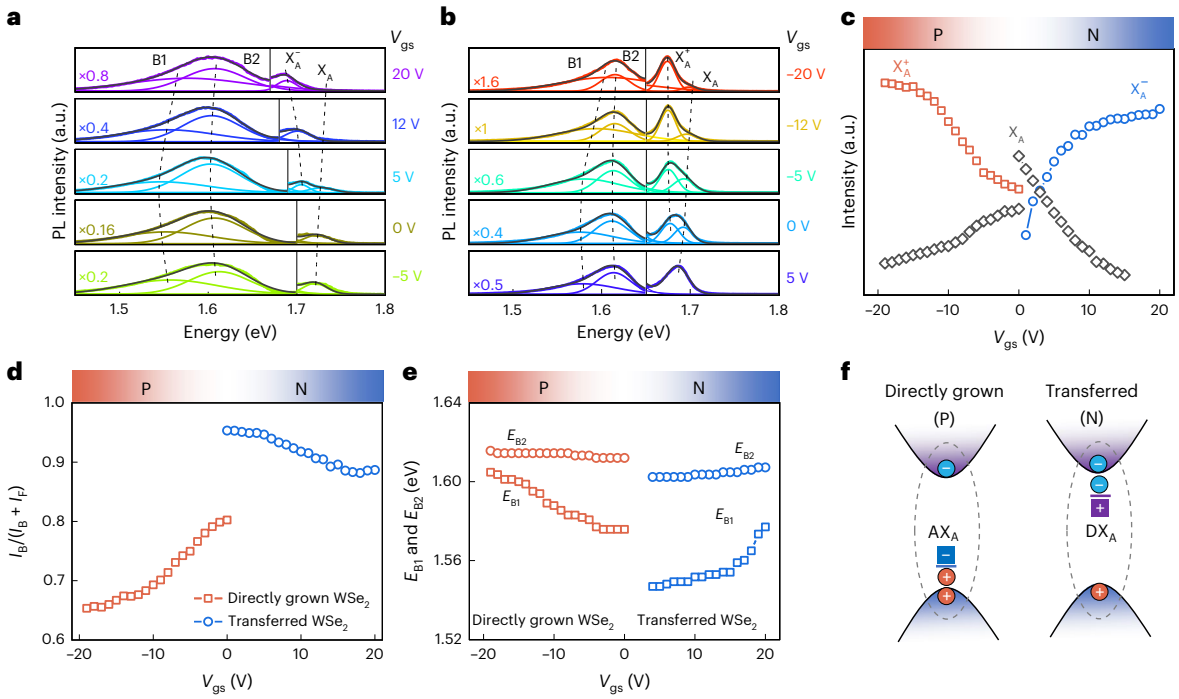


Fig. 2 | Gate-dependent PL of monolayer WSe_2 at 3.4 K. a,**b**, PL spectra of WSe_2 transferred from sapphire (**a**) and WSe_2 directly grown on SiO_2/Si (**b**) at 3.4 K with different gate voltages (V_{gs}). Each spectrum is fitted with four Gaussian peaks, denoted as B1, B2, X_A and X_A^- (or X_A^+). **c**, Trion and exciton peak intensity versus gate voltage. **d**, Percentage of bound-exciton-associated accumulative PL intensity ($I_B/(I_B + I_F)$) as a function of V_{gs}. **e**, Peak positions E_{B1} and E_{B2} as a function of V_{gs}. **f**, Schematic representing different origins of bound excitons in the two samples.

different gate voltages (V_{gs}) are plotted in Fig. 2a,b (also shown in Supplementary Fig. 11). The PL emission of free excitons at higher photon energies (ranging from 1.65 to 1.75 eV) can be deconvoluted by two Gaussian functions. These functions are assigned to charged and neutral A excitons, respectively (X_A^- and X_A in Fig. 2a and X_A^+ and X_A in Fig. 2b). In the transferred WSe_2 samples, as V_{gs} increases from 0 V to 20 V (more electrons are doped into WSe_2), the X_A resonance energy moves to higher energy and its oscillator strength (Fig. 2c) diminishes as the X_A^- resonance develops a redshift due to Coulomb screening, suggesting that these samples are initially n doped. By contrast, the as-grown WSe_2 samples exhibit a similar trend for the X_A and the X_A^+ resonances when V_{gs} decreases from 0 V to −20 V, indicating that these samples are p doped at $V_{gs} = 0$. Similar behaviours have also been observed in gated high-quality exfoliated samples[25,26]. To further elucidate the nature of the bound excitons, we plot $I_B/(I_B + I_F)$ as a function of V_{gs} (Fig. 2d), where I_B and I_F are the accumulated PL intensities for bound excitons and free excitons, respectively. The intensities for both samples decrease with increasing carrier density, which is attributed to a Coulomb screening effect[27]. In addition, the I_B value of the transferred WSe_2 samples is greater than that of the as-grown WSe_2 samples, indicating a high defect state density. The gate-dependent peak energies of the B1 and B2 bound exciton emissions are shown in Fig. 2e. The resonance energies of the B2 bound excitons for both types of sample show weak gate dependence, which is related to structural defects[14,16,26]. The B1 peaks show the same trend when the carrier density increases, which can be attributed to exciton recoil effects[26]. Selenium vacancies (V_{Se}) have been identified as the predominant point defects in CVD-grown monolayer WSe_2, independent of the substrate[28]. These vacancies can be healed by means of oxygen passivation. In addition, V_{Se} has been reported as the primary contributor to the n-type conductivity in unintentionally doped WSe_2, whereas the p-type conductivity in unintentionally doped WSe_2 is still under debate and has been attributed to tungsten vacancies (V_W) or to complex defects (O-V_W or O-V_{Se}) with oxygen (given that oxygen is present in the growth conditions)[29–33]. We identified the B1 state in the transferred WSe_2 film as a V_{Se}-induced donor-type bound exciton state, which experiences a blueshift in resonance energy when external electrons are introduced. By contrast, the B1 state in the directly grown WSe_2 film is an acceptor-type bound exciton state, whose origin can be attributed to the combined contribution from spotted V_W or to the formation of complex O-V_W or O-V_{Se} defects, with its resonance energy also exhibiting a blueshift as external holes are introduced.

Improved performance of monolayer WSe_2 pFETs with oxygen passivation

To decrease the defect density in monolayer WSe_2, we developed an industry-compatible and tunable O_2 annealing process that tunes the performance of monolayer WSe_2 transistors by varying the O_2 flow to tune the vacuum level in the furnace before annealing. Figure 3a shows the low-temperature PL measurements of pristine WSe_2 and WSe_2 annealed in O_2 at 430 mtorr (O_2-WSe_2) (Supplementary Fig. 12 shows WSe_2 annealed at 26 mtorr (O-WSe_2)). The bound-exciton-associated accumulative PL intensities ($I_B/(I_B + I_F)$) for the pristine and O_2-WSe_2 samples are 0.95 and 0.65, respectively. Therefore, the lower intensities of the B1 and B2 peaks for O_2-WSe_2 corroborate a lower defect density after O_2 annealing at 430 mtorr. Figure 3b,c shows the transfer and output characteristics of a 2-μm-long O_2-WSe_2 pFET with an ultrahigh on/off of 10^{10} at a V_{ds} of −2 V. The μ_{FE} value at room temperature is 137 $cm^2 V^{-1} s^{-1}$, which is three times larger than that of pristine WSe_2 (Supplementary Fig. 13). We also present the statistical distribution of the μ_{FE} (Fig. 3d) and average μ_{FE} values of pristine and O_2-WSe_2 pFETs, which are 40 and 122 $cm^2 V^{-1} s^{-1}$, respectively (Supplementary Fig. 14). The O_2 annealing process was also applied to the transferred monolayer WSe_2 transistor, which exhibited improved p-type transport capacity after O_2 annealing (Supplementary Fig. 15). Moreover, the drain current of monolayer WSe_2 decreases by approximately two times after annealing at 200 °C in Ar, indicating that oxygen plays an important role in passivating V_{Se} (Supplementary Fig. 16). High-angle annular dark-field scanning transmission electron microscopy

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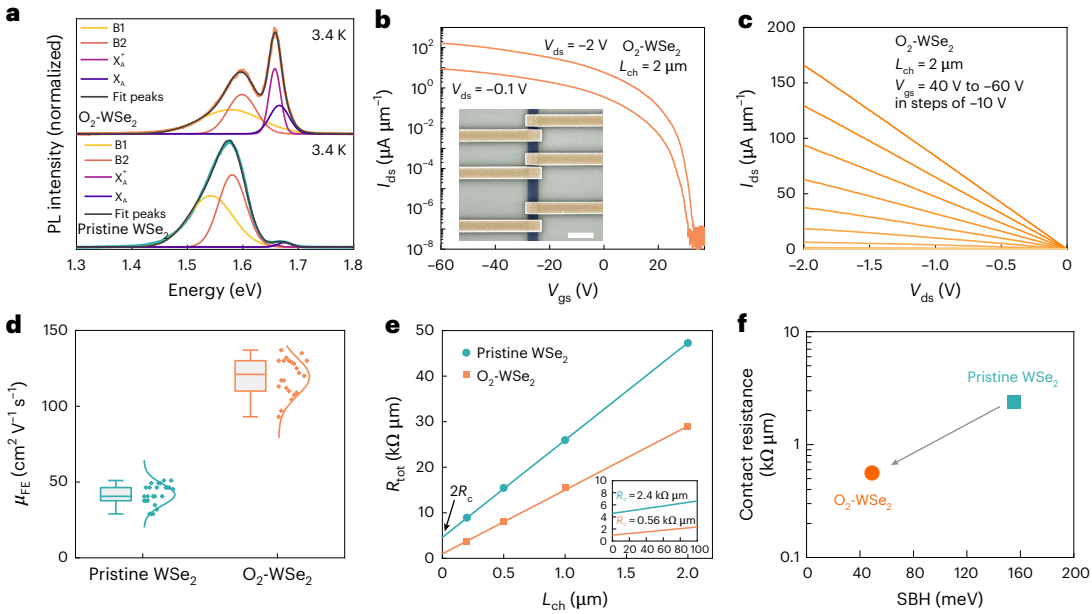


Fig. 3 | PL and electrical characterization of monolayer WSe_2 on SiO_2 before and after O_2 annealing. a, PL spectra of monolayer WSe_2 grown on SiO_2 substrates with and without O_2 annealing at 3.4 K. The fitted peaks can be assigned to neutral excitons (X_A), trions (X_A^+) and defect-bound excitons (B1 and B2). **b**,**c**, Transfer (**b**) and output (**c**) characteristics of monolayer WSe_2 devices annealed in O_2 at 430 mtorr. The gate dielectric is 100-nm SiO_2. The channel length and width are 2 and 1 μm, respectively. The inset shows an SEM image of a WSe_2 transmission-line method structure with a minimum L_{ch} of 100 nm. Scale bar, 2 μm. **d**, Distribution of the hole field-effect mobility (μ_{FE}) of monolayer WSe_2 before and after O_2 annealing at 430 mtorr. In the box plot, the box represents the interquartile range from the 25th to the 75th percentile, and the line inside the box denotes the median. The whiskers extend to the minimum and maximum values, excluding outliers. For pristine WSe_2, n = 25. The minima, maxima and centre values are 29, 51 and 40, respectively. For O_2-WSe_2, n = 22. The minima, maxima and centre values are 93, 137 and 122, respectively. **e**, Transmission-line method resistances of monolayer WSe_2 pFETs before and after O_2 annealing at 430 mtorr. After annealing, R_c is determined to be 0.56 kΩ μm at $n_s \approx 1.1 \times 10^{13}$ cm^{-2}, which is four times lower than that of the pristine device. The channel lengths are 200 nm, 500 nm, 1 μm and 2 μm. **f**, R_c versus SBH. The R_c values are extracted from **e**.

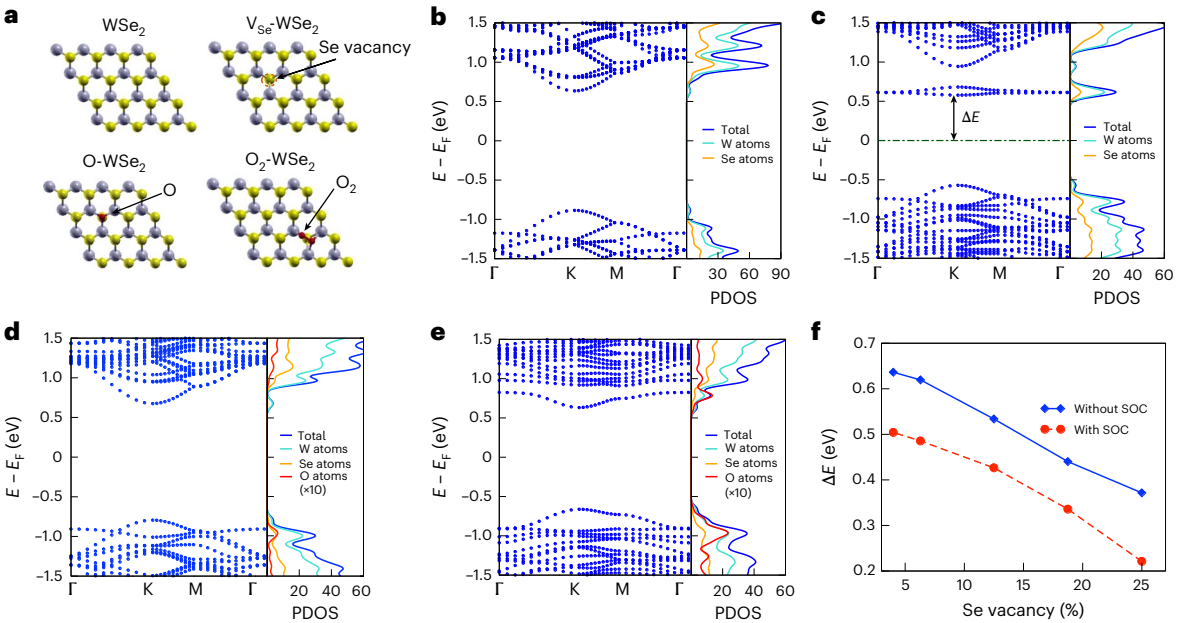


Fig. 4 | First-principle theoretical simulations of pristine monolayer WSe_2, with V_{Se}, O and O_2 passivation. a, Crystal structure of a 4 × 4 supercell of pristine WSe_2 (top left), with a single Se vacancy (top right), an O-passivated vacancy (bottom left) and an O_2-passivated vacancy (bottom right). **b**–**e**, Band structure and projected density of states (PDOS) without SOC for pristine (**b**), single selenium vacancy (**c**; including the definition of the energy difference between the minimum unoccupied defect state and the Fermi energy level (ΔE) as used in the main text), single selenium vacancy passivated with O (**d**) and with O_2 (**e**). **f**, ΔE as a function of the selenium vacancy concentration with and without SOC.

(STEM) also revealed that selenium vacancies can be passivated by O_2 annealing (Supplementary Fig. 17)[34]. We also did not observe the formation of WO_x in terms of Raman, X-ray photoelectron spectroscopy or transmission electron microscopy (TEM) characterization (Supplementary Figs. 18 and 19). Higher-temperature annealing causes the formation of WO_x and the degradation of drain current and subthreshold swing (Supplementary Fig. 20). Therefore, the higher hole mobility in O_2-WSe_2 is mainly attributed to the effective passivation of

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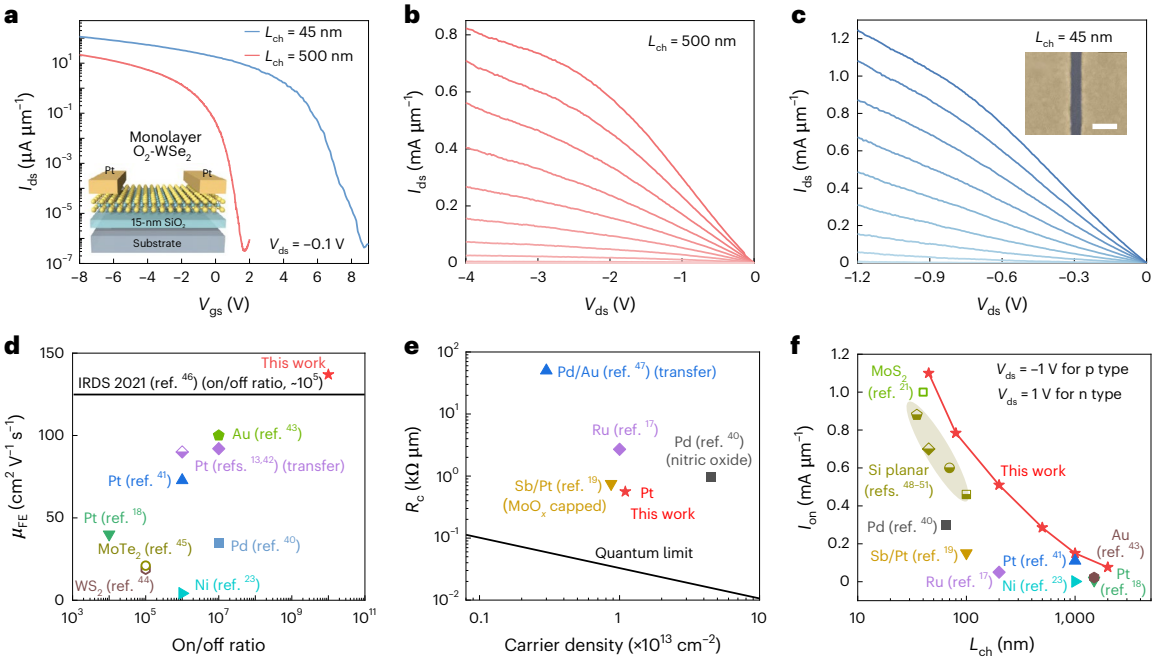


Fig. 5 | Device characterization of short-channel monolayer WSe_2 pFETs. **a**, Transfer characteristics of WSe_2 devices with channel lengths of 45 nm and 500 nm at 300 K. Inset: schematic of a monolayer WSe_2 device annealed with O_2 at 430 mtorr in this work. **b**, Corresponding output characteristics of WSe_2 pFETs in **a** with L_{ch} of 500 nm. V_{gs} ranges from 2 to −9 V with steps of −1 V. **c**, Output characteristics of WSe_2 pFETs in **a** with an L_{ch} of 45 nm. V_{gs} ranges from 9 to −9 V with steps of −2 V. The inset shows an SEM image of a WSe_2 transistor with an L_{ch} of 45 nm. Scale bar, 100 nm. **d**, Summary of the on/off ratio and hole mobility for CVD monolayer WSe_2 transistors reported in the literature. The requirements of IRDS 2021 and other typical p-type transition metal dichalcogenide materials (WS_2 and $MoTe_2$) have also been added for comparison. **e**,**f**, Benchmarks of the contact resistance R_c (**e**) and on-state current I_{on} (**f**) for CVD monolayer p-type WSe_2 transistors at room temperature. The black line in **e** represents the quantum limit of R_c. The reported highest I_{on} for monolayer MoS_2 nFETs (dark green with open symbols) at L_{ch} = 40 nm with V_{ds} = 1 V and p-type silicon technology (deep yellow with half-filled symbols) have also been added for comparison.

Se vacancies and a decrease in the defect density in the WSe_2 film by oxygen incorporation[34,35]. The corresponding contact resistance R_c is extracted using the transmission-line method structure, and the results are shown in Fig. 3e. The extracted R_c value of pristine WSe_2 is 2.4 kΩ μm, which can be reduced to 0.56 kΩ μm for the O_2-WSe_2 pFET at a carrier density $n_s = 1.1 \times 10^{13}$ cm^{-2}. To investigate the SBHs of monolayer WSe_2 devices, a thermionic emission model was applied to extract the Schottky barriers at the Pt/WSe_2 junction (Supplementary Fig. 21)[36]. The SBH in the O_2-WSe_2 FET under flat-band conditions is 49 meV, which is much lower than that extracted from pristine WSe_2 devices (155 meV). The decrease in the SBH caused by O_2 annealing was further confirmed by theoretical calculations (Supplementary Figs. 22–24). Figure 3f shows the SBH versus R_c for monolayer WSe_2 with and without O_2 annealing. As expected, O_2-WSe_2 FETs exhibit a low R_c of approximately 0.56 kΩ μm owing to their low SBH.

Theoretical analysis of monolayer WSe_2 with Se vacancies and oxygen passivation

To provide physical insights into the experimental results, we performed ab initio density functional theory using the QUANTUM ESPRESSO suite[37]. Specifically, we focused on the effects of V_{Se} and oxygen (O and O_2) passivation of monolayer WSe_2. In Fig. 4, we report the relaxed crystal structure of all the studied geometries (Fig. 4a), band structure and projected density of states for pristine WSe_2 (Fig. 4b), in the presence of a single selenium vacancy (Fig. 4c) and the case in which the selenium vacancy is passivated with O (Fig. 4d) or O_2 (Fig. 4e) for a 4 × 4 WSe_2 supercell. We clearly observe that V_{Se}-WSe_2 produces a defective state that disappears when the vacancy is passivated with O or O_2, in agreement with the experimental findings described above. In addition, we observe from the projected density of states that in the O-WSe_2 and O_2-WSe_2 structures, a state appears close to the top of the valence band, which is more pronounced in the O_2-WSe_2 structure, and can indicate p-type doping, as observed experimentally. We also note that the presence of oxygen, in addition to removing the defects state close to the bottom of the conduction band, changes the band structure with respect to that of pristine WSe_2. This difference is more pronounced for O_2-WSe_2 but also for O-WSe_2 and can motivate the change in the transport properties observed in O-WSe_2 pFETs[27].

We also studied the impact of different V_{Se} concentrations to understand the different (n or p) types of behaviour observed experimentally. To do so, we considered V_{Se} concentrations of 4%, 6.3%, 12.5%, 18.8% and 25% obtained by including a single vacancy in a 5 × 5 (4%) and a 4 × 4 (6.3%) supercell and two (12.5%), three (18.8%) and four (25%) vacancies in a 4 × 4 supercell (Supplementary Fig. 25)[38]. We performed two separate sets of calculation with and without spin–orbit coupling (SOC). In Fig. 4f, we report the energy difference between the minimum unoccupied defect state and the Fermi energy level ($\Delta E = E_{\text{defect state min}} - E_F$), as defined in Fig. 4c. A smaller ΔE indicates an n-type behaviour, whereas larger values indicate a p-type behaviour. In Fig. 4f, we clearly observe that as the density of defects increases, ΔE decreases, leading to a switch from p-type (with no-defect or low-density Se vacancies) to n-type behaviour, both in the case with and without SOC. We finally observe, as expected, a smaller value of ΔE for the case with SOC, due to the splitting of the defect state[39].

High-performance short-channel monolayer O_2-WSe_2 pFETs

To demonstrate the performance potential of monolayer O_2-WSe_2 transistors, channel length scaling was performed using 15-nm SiO_2 as the gate dielectric (Fig. 5a, inset). Figure 5a shows the transfer characteristics of monolayer O_2-WSe_2 pFETs with L_{ch} values of 45 nm and 500 nm. Both devices exhibit high on/off ratios due to the atomically thin channel materials. Moreover, the 45-nm-long short-channel device has an excellent on/off ratio of nearly 10^9 at V_{ds} = −0.1 V, indicating its

promising application in low-power electronics. Moreover, O_2 annealing can also reduce the subthreshold swing and drain-induced barrier lowering of WSe_2 transistors (Supplementary Fig. 26), which can be further improved by decreasing the thickness of the gate dielectrics (Supplementary Fig. 27). In addition, a shift in threshold voltage towards a positive voltage for the 45-nm-long device was observed because of the contact-doping effect (Supplementary Fig. 28). Moreover, the device with O_2 annealing exhibited excellent stability (Supplementary Figs. 29 and 30). Figure 5b shows the corresponding output characteristics of the 500-nm short-channel transistor, which exhibited a high drain current of 823 μA $μm^{-1}$ at $V_{gs} = -9$ V and $V_{ds} = -4$ V. With scaling down to 45 nm, the maximum I_{on} at $V_{gs} = -9$ V and $V_{ds} = -1.2$ V increased from 367 μA $μm^{-1}$ to 1,245 μA $μm^{-1}$ (Fig. 5c) without the observation of a self-heating effect in the high-field transport[21], which is contrary to that of n-type MoS_2. Figure 5d shows the benchmark $μ_{FE}$ as a function of the on/off ratio compared with previous studies on p-type monolayer WSe_2 (Supplementary Table 1)[13,18,23,40–45] (https://irds.ieee.org/editions/2021/more-moore). It is clear that the O_2-WSe_2 transistor in this work concomitantly achieves both the highest $μ_{FE}$ and on/off ratio, indicating promising applications for ultimate CMOS scaling with high-performance low-power electronic transistors. We also benchmarked our transistor performance with R_c, a key parameter limiting the scaling of I_{on} (Fig. 5e)[17,19,40,46]. Compared with the other p-type doping methods, a low R_c of ~560 Ω μm is achieved through a standard CMOS-compatible annealing process, which promises good reliability and scalability. We compare the I_{on} value in this work with other monolayer WSe_2 pFETs reported in the literature at the same $V_{ds} = -1$ V and maximum V_{gs} (Fig. 5f and Supplementary Fig. 31)[17–19,21,23,40,41]. Due to the reduction in R_c, our approach offers the highest performance at various L_{ch} values compared with the values reported in the literature (Supplementary Table 1). Moreover, the I_{on} value of monolayer WSe_2 pFETs is greater than that of p-type Si transistors at a similar technology node[47–50] and satisfies the 2028 roadmap target for high-performance logic transistors[21] (https://irds.ieee.org/editions/2021/more-moore). The I_{on} value of a monolayer WSe_2 pFET at an L_{ch} of 45 nm in this work is also higher than that of previously reported n-type monolayer MoS_2 (ref. 21). The results indicate that oxygen doping by annealing under vacuum conditions is an effective and reliable way to realize high-performance WSe_2 pFETs with low R_c and high I_{on}.

Conclusion

We have reported an industry-compatible low-temperature oxygen-incorporated technology that can passivate selenide vacancies and suppress defect states in CVD monolayer WSe_2. The incorporation of oxygen not only increases the hole carrier mobility (~137 $cm^2 V^{-1} s^{-1}$) and decreases the SBH and contact resistance. A drain current of 1,245 μA $μm^{-1}$ at $V_{ds} = -1.2$ V is achieved in p-type monolayer WSe_2, which outperforms equivalent silicon CMOS technologies and satisfies the 2028 roadmap target. In addition to p-type WSe_2, we also demonstrated high-performance p-type molybdenum ditelluride ($MoTe_2$) devices (Supplementary Fig. 32), suggesting that low-temperature oxygen-incorporated technology may be a general doping technology for creating high-performance p-type transistors for CMOS applications.

Methods

Growth of monolayer WSe_2

WSe_2 single crystals were grown by a molten-salt-assisted low-pressure CVD process. First, Se (Sigma-Aldrich, ≥99.5%) was prepared in a ceramic boat, WO_2 (Alfa Aesar, 99.99%) and KCl (Sigma-Aldrich, 99.999%) were placed in a corundum boat with SiO_2/Si and sapphire substrates, as described above. Then, the precursors were placed in a vacuum one-inch tube furnace. Finally, the crystals were grown at 890 °C for 10 min under low pressure (1,700 Pa) with a mixture of 100 standard cubic centimetres per minute (s.c.c.m.) of Ar and 9 s.c.c.m. of H_2 as the carrier gas. After growth, the furnace was naturally cooled to room temperature.

Transfer of monolayer WSe_2

After CVD growth, monolayer WSe_2 crystals on sapphire were transferred onto a target substrate using a wet transfer process. Polymethyl methacrylate (PMMA) was first spin coated onto the as-grown WSe_2/sapphire as a protection layer. Then, a thermal release tape (TRT) was applied to the PMMA/WSe_2/sapphire to avoid possible folding during the transfer process. The TRT/PMMA/WSe_2/sapphire stacked film was immersed in deionized water for 4 h. Next, the TRT/PMMA/WSe_2 stacked film was placed on the target substrate; the TRT was released by baking on a hotplate at 100 °C; the PMMA film was finally removed by immersing the sample in acetone for 12 h, leaving behind monolayer WSe_2 crystals on the target substrate.

Preparation of STEM sample

The STEM samples were prepared with a wet-etching transfer method. The PMMA was spin coated on a SiO_2/Si substrate deposited with monolayer WSe_2 flakes. The obtained substrate was then immersed in a KOH solution (2 M) and deionized water sequentially to etch the SiO_2 layers and remove the KOH residue. Afterwards, a film floated on the water and was fished by a TEM grid to obtain a PMMA/WSe_2/grid structure. After drying overnight in a nitrogen cabinet, the mentioned structure was dropped into acetone and then annealed (with or without O_2) at 250 °C under vacuum for 2 h to remove the PMMA.

TEM and STEM

Cross-sectional TEM lamellas of the FET samples were prepared using a Thermo Scientific Helios G4 HX dual-beam focused ion beam/scanning electron microscopy (SEM) machine using in situ focused ion beam lift-out technology. Before extraction and thinning, we coated the surface of the sample with a carbon film to protect the sample surface from ion beam damage. High-angle annular dark-field STEM and cross-sectional TEM were performed using a Thermo Scientific Themis *Z*-aberration-corrected transmission electron microscope or a Thermo Scientific Tecnai F20 at an accelerating voltage of 200 kV. EDS mapping were obtained using the Super X FEI System in the STEM mode. All the above characterization methods were performed by Eurofins EAG Materials Science China.

Material characterization

Monolayer WSe_2 was confirmed using Raman and PL spectra measurements at a laser wavelength of 532 nm. Raman spectroscopy was obtained through inVia Reflex manufactured by Renishaw. For temperature-dependent PL, a Montana Instruments Cryo Advance was used to control the temperature and a Keithley 2450 device was used to provide the gate voltage. The test light path was custom made and all the PL spectra were acquired in a vacuum environment. Atomic force microscopy was carried out with an SPM-9700 device in the dynamic mode. The SEM images were obtained with a Raith 150-Two instrument. X-ray photoelectron spectroscopy was performed at the Huazhong University of Science and Technology Analytical & Testing Center using an AXIS SUPRA+.

Device fabrication and measurement

For back-gate device fabrication, both WSe_2 materials grown directly on SiO_2 and transferred from sapphire were patterned by electron-beam lithography. AR-P 6200.13 electron-beam resists are used during the plasma etching of WSe_2 films. Then, the channel was isolated by reactive ion etching (SHL 100μ-RIE) with 100/10/80-s.c.c.m. flow of $Ar/O_2/CF_4$ gas at 3 W for 180 s followed by immersing the sample in *N*-methyl pyrrolidone for 15 h to remove the electron-beam resist AR-P 6200.13. After annealing at 250 °C for 2 h with 100-s.c.c.m. Ar to remove the AR-P residuals, source/drain contacts were defined by

electron-beam lithography on the PMMA A4 resist and formed with metal deposition by electron-beam evaporation. After the metal was lifted from unwanted regions with acetone, the p-type WSe_2 devices were annealed in an O_2 atmosphere under different vacuum conditions for 2 h at 200 °C. Electrical characteristic measurements were characterized at a Lakeshore probe station at room temperature under vacuum (<10^{-5} torr) using an Agilent B1500A semiconductor parameter analyser.

Theoretical methods

Density functional theory package in the Atomistix ToolKit is used for the calculation of geometric optimization and band alignment properties[51]. The Perdew–Burke–Ernzerhof approach under the generalized gradient approximation is used to describe the exchange–correlation interaction[52]. The SG15 pseudopotential is used as the ultrabasis set with a density mesh cut-off of 90 Hartree. For geometric optimization, the convergence criteria of energy and force are 10^{-5} eV and 0.02 eV Å^{-1}, respectively. The Brillouin-zone integrations are sampled with a $6 \times 6 \times 1$ *k*-point mesh. A vacuum layer of 40 Å is used to eliminate the interactions between periodic boundaries. For calculations with SOC, we use SOGGA exchange–correlation interaction with the SG15-SO pseudopotential. The DFT-D3 method is used for the analysis of vertical contact systems[53].

To study the effect of V_{Se} and oxygen (O and O_2) passivation of monolayer WSe_2, density functional theory calculations have been performed with the QUANTUM ESPRESSO suite with and without SOC[34]. We used projector augmented-wave pseudopotentials within the Perdew–Burke–Ernzerhof approximation for the exchange–correlation functional. All the structures considered were geometrically relaxed until all the forces were smaller than 0.051 eV Å^{-1}. For the electronic structure calculations, we considered an energy convergence threshold of 10^{-6} Ry, and an $8 \times 8 \times 1$ Γ-centred *k*-mesh grid.

Data availability

The data that support the plots within this paper are available from the corresponding authors upon reasonable request.

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Author contributions

G.F., Xuefei Li and X.M. proposed and supervised the project. L.S. and T.G. fabricated and measured the devices. Xuefei Li and Y.H. analysed the results and experimental plans. T.C., D.M., G.I., G.F., J.Y. and S.Z. performed and discussed the first-principles calculations. Xiubin Li and J.L. performed the low-temperature PL characterization and analysed the results. L.S. and T.G. performed the X-ray photoelectron spectroscopy, SEM, atomic force microscopy, PL and Raman characterizations. All authors discussed the results and wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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